

Zhengyang Yu

Curriculum Vitae

San Diego & Bay Area

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA

Expected Graduation: June 2028

Relevant Coursework: Advanced Data Structures, Design and Analysis of Algorithms, Computer Organization, Machine Learning, Systems Programming

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*

- Designed and built a network-connected ESP32-S3 system in C++ that ingests and visualizes live quota data across multiple accounts, with touch interaction, reset countdowns, weather, and alerts.
- Implemented concurrent FreeRTOS tasks, adaptive HTTPS polling, PSRAM-aware rendering, device-code authorization, and bounded persistent logging with runtime diagnostics under memory and API-rate constraints.

2025 **HTML to ePub Converter**, *Personal Project*

- Engineered a converter to transform HTML content into ePub format, implementing scalable parsing and metadata embedding.
- Enhanced reliability and efficiency through structured error handling and modular component design.

2024 **Home Assistant Presence Plugin**, *Personal Project*

- Built a Python Home Assistant integration that derives real-time room occupancy from entry and exit sensor events.
- Designed an event-driven pipeline using REST APIs to maintain occupancy state and automate smart lighting reliably.

Skills

Programming C++, Python, Java, C

Systems Linux, Docker, Git, REST APIs, HTTPS, FreeRTOS, concurrent and event-driven systems

AI-Assisted Codex-assisted implementation, refactoring, debugging, code validation, and technical documentation

Concepts Data Structures, Algorithms, Object-Oriented Design, Software Design, Testing, Reliability, Debugging

Platforms ESP32-S3, PlatformIO, touch and IMU integration, VS Code

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch

Supported community members with computer troubleshooting and taught basic software skills.

2024 Tutor, CTN (Community Technology Network)

Provided one-on-one tutoring to help adults improve digital literacy skills, including computer operation and internet usage.